



DEPARTMENT OF
**SPACE, PLANETARY AND
ASTRONOMICAL SCIENCES
AND ENGINEERING**

Indian Institute of Technology Kanpur



2026-27 | Placement Brochure

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ABOUT US

The Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE) at IIT Kanpur is dedicated to advanced education and research in space science, planetary exploration, and astronomy. It brings together cutting-edge work in instrumentation, observations, data analysis, and theoretical modeling.

SPASE offers postgraduate and doctoral programs, along with a minor for undergraduates, and promotes interdisciplinary learning at the intersection of science and engineering. Key research areas include planetary science, astrophysics, multi-wavelength astronomy, remote sensing, and space instrumentation.

The department also actively engages in public outreach to foster awareness and enthusiasm for space and astronomy among students and the general public.

HEAD OF THE DEPARTMENT



Prof. Amitesh Omar
Head of Department

The Space Planetary and Astronomical Sciences & Engineering (SPASE) is among the newest departments at IIT Kanpur. The department extends a warm welcome to you. Presently, PG programs are offered in the department. Our teaching methodology incorporates experiments and hands-on experience without sacrificing the rigour of the subject. The department's curriculum is multidisciplinary. Students are introduced to fundamental concepts such as fluid mechanics, radiative processes, and orbital dynamics, trained to understand optical, radio, and X-ray instruments, and the analysis of data relevant to space science & technology. Our students join us with diverse academic backgrounds with UG/PG degrees in computer science, electrical engineering, mechanical engineering, physics, mathematics, and geophysics. Our faculty members have led major projects of national and international importance and are collaborating with startups, industries and various government organizations like DST, ISRO, and DRDO. In presenting our students to your esteemed organization, I am convinced that you will find them to be well-prepared, brimming with enthusiasm, and capable of becoming future leaders.

FACULTY PLACEMENT COORDINATOR



Prof. Mohit Bhardwaj
Faculty Placement Coordinator

Few industries are evolving as rapidly as the global space sector. From satellite communications and Earth observation to autonomous systems, scientific instrumentation, planetary exploration, and commercial spaceflight, innovation increasingly occurs at the intersection of multiple disciplines. Organizations today seek graduates who not only possess strong technical foundations, but also the ability to integrate scientific understanding, engineering design, and computational thinking to solve complex technological challenges.

The Department of Space, Planetary and Astronomical Sciences & Engineering (SPASE) at IIT Kanpur was founded on this very philosophy. By bringing together the scientific perspectives of astronomy and planetary science with the engineering principles that enable modern space technologies, SPASE offers a distinctive interdisciplinary environment where curiosity drives innovation and ideas translate into solutions. Our students are encouraged not merely to master existing knowledge, but to question assumptions, connect disciplines, and address real-world challenges with creativity and rigor.

Through research-led learning, hands-on engineering, and multidisciplinary collaboration, our graduates develop the analytical depth, practical experience, and professional maturity required to excel across aerospace, defense, geospatial technologies, scientific instrumentation, semiconductor technologies, artificial intelligence, and the rapidly expanding new-space ecosystem. They graduate not only with technical expertise but also with the adaptability, initiative, and systems perspective needed to contribute effectively from day one and grow as future technology leaders.

As the Department Placement Coordinator, I view recruitment as the beginning of a lasting partnership. On behalf of the SPASE community, I warmly invite you to engage with our students, explore the breadth of talent they represent, and discover individuals who are prepared to create meaningful impact in your organization and beyond.



DEGREES OFFERED AND DEMOGRAPHICS

01

MINOR FOR UG

02

M.TECH

03

PHD

Computer Science Engineering

M.Sc Physics

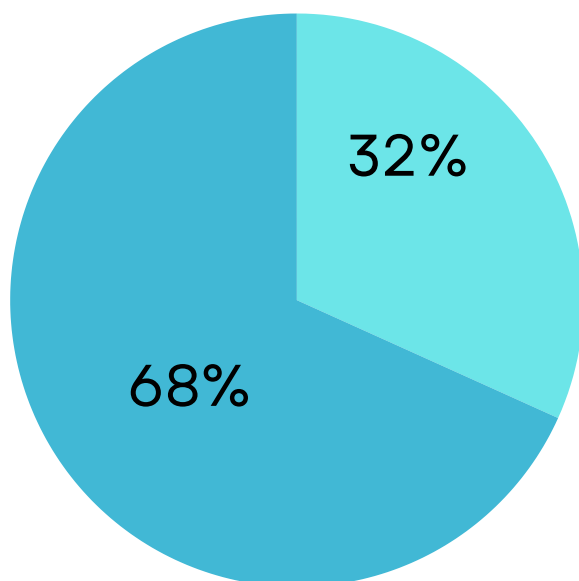
Information and Technology

Mechatronics

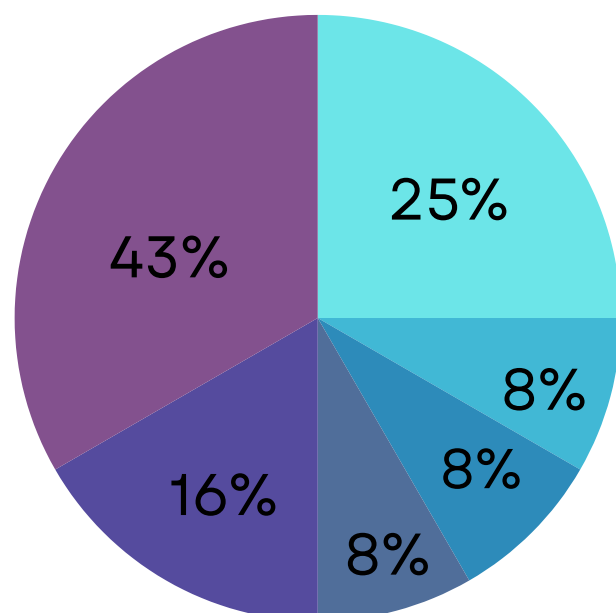
Mechanical Engineering

Aerospace Engineering

M.Tech Students PHD Students



Degree Demographics



Academic Background M.Tech Y25

ACADEMIC COURSE LIST

UG Courses

- Introduction to Astronomy and Astrophysics
- Galaxies and Observational Cosmology
- Introduction to Man-made Satellite Systems and its Environment
- Observing the Milky Way Galaxy - Our Home in the Cosmos
- Introduction to Space Technology
- Space Economics - Law, Policy and Benefits

PG Courses

- Introduction to space & astronomical instrumentation
- Solar system processes and space missions
- Fundamentals of Astronomy and Astrophysics
- Radio Astronomy
- Introduction to Celestial Observational Techniques
- Introduction to Celestial Mechanics
- Stars and Stellar Evolution
- Space Instrumentation Laboratory
- Introduction to radiative processes in space
- Planetary Surface Processes
- Mathematical Techniques in Space Sciences & Engineering
- Numerical Techniques in Space Sciences & Engineering
- Optical Instrumentation Laboratory
- Space environment and Space systems
- Introduction to fluid mechanics in space
- Statistical Methods and Applications in Space Sciences
- Introduction to Mathematical Transforms and Applications in Space Sciences
- Introduction to Astrobiology
- Applied Optics: From Rays to Waves
- Radio instrumentation, observing techniques, and data analyses (Radio Lab)
- Spacecraft Dynamics
- High Energy Astrophysics of Binary Star Systems

Additional Courses

- Introduction to Machine Learning
- Human Computer Interaction
- Aerospace, Flapping Wings

SOFTWARE & UTILITIES



Artemis Capture



LAXPC Software





LABS & RESEARCH FACILITIES

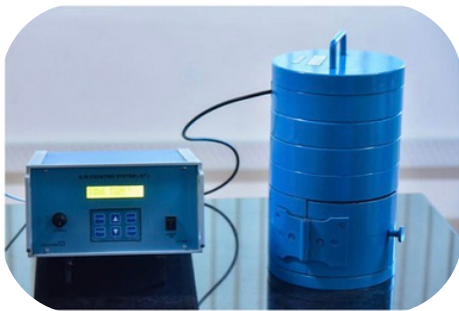
Space Instrumentation Laboratory

The course provided hands-on experience on instrumentation techniques related to astronomy, planetary and space sciences and engineering.



Study of Compact Object using Real Space Data (Astrosat)

Students used data of LAXPC instrument. This data was used to drive basic physical properties of the black hole and neutron star X-ray binaries like temperature, inner disk radii, source energy spectrum state: soft or hard, spectrum timing properties, thermonuclear burst, neutron star spin frequency, flux variation and others.



Optical Instrumentation Laboratory

The course provided comprehensive knowledge on instrumentation techniques related to astronomy, planetary and space sciences and engineering.



Radio Instrumentation, Observing Techniques and Data Analysis

This course provided comprehensive knowledge on instrumentation, observing techniques, and data analyses related to radio astronomy via tutorials and lab experiments.



OUR FACULTY MEMBERS



Research Interests
Galaxy astrophysics, Radio & Optical astronomy instrumentation

Amitesh Omar



Research Interests:
Astrophysical transients, jets, compact objects, Galactic Center, Life in the Universe

Kunal P. Mooley



Research Interests
Planetary science, Granular Mechanics, Contact Mechanics, Applied Mechanics

Ishan Sharma



Research Interests
Radio Astronomy, Interferometry, Astronomical Instrumentation and Space Weather

Mugundhan V.



Research Interests
Solar Physics, Space Weather, Radio Astronomy, Plasma Physics, Radio Wave Propagation, Data Science, Imaging Algorithms

Rohit Sharma



Research Interests
Characterization of Exoplanets: direct imaging, transmission spectroscopy. Adaptive optics & wavefront control techniques. Ground & space-based optical & infrared instrumentation.

Prashant Pathak



Research Interests
Fast Radio Bursts, Compact Objects, Astrophysical Transients, Radio Astronomy Instrumentation, Astrostatistics

Mohit Bhardwaj



Research Interests
Cosmology, non-linear large scale structure, non-linear dynamics.

Sharvari Nadkarni Ghosh



Research Interests
Geodynamics, Interior rheology of planets, Onset of convection and thermal evolution of planets, Geophysical inversion

Chhavi Jain



Research Interests
Space Weather interaction of Spacecraft, Computational Electromagnetics, Design and development of Antennas for Radio Telescopes and Microwave Sensors

Soumyabrata Chakrabarty
Professor of Practice

**Research Interests**

Cosmology, Astrophysics,
Cosmic Rays, X-ray
Astronomy, Nuclear Fusion
at low energies

Pankaj Jain

Emeritus professor

**Research Interests**

Galactic and Extragalactic
Astrophysics (Theory,
Idealised/Cosmological
Simulations & Observations)

Sandeep Kumar Kataria

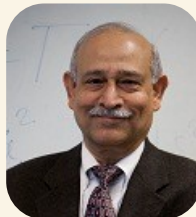
DST inspire faculty

**Research Interests**

Granular Matter, soft impact
dynamics, planetary
cratering

Hiroaki Katsuragi

Distinguished visiting professor

**Research Interests**

Satellite dynamics and
control, space robotics, and
dynamics of aerospace
structures

Arun Misra

Distinguished visiting professor

**Research Interests**

Planetary Science, Orbital
Dynamics

Renu Malhotra

Distinguished visiting professor

**Research Interests**

Space Policy & Regulations,
Space Economy, Industrial
AI, Commercialization &
Impact Pathways

Abhay Gupta

Distinguished visiting professor

**Research Interests**

Astrobiology, Microbe-
Mineral Interaction,
Microbial Ecology

Yamini Jangir

Visiting faculty

MASTERS STUDENTS Y25



Amey

Research Interests

Exoplanets
Atmospheres,
Planetary interiors



Anirudh Singh

Research Interests

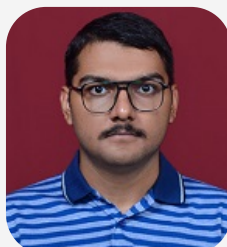
Optical-Mechanical
Design &
Instrumentation



Bahniman Ghosh

Research Interests

Space Instrumentation:
FPGA-based Digital
Signal Processing for
Solar Radio
Observations



Diptangshu Dey

Research Interests

Space Debris
tracking, Space
Robotics, Space
Instrumentation.



Intakhab Alam

Research Interests

Intelligent Astronomical
Software for Multi-
wavelength and Multi-
messenger Transient
Astronomy



Pranay Shukla

Research Interests

Applied Machine
Learning, Data
Analytics, Intelligent
Control Systems, and
Optimization.



Rohit Kumar

Research Interests

Space manufacturing:
Developing binders
for Lunar Regolith



Saurav Mishra

Research Interests

Computational Modeling
of Heliospheric
Environments,
Ionospheric Prediction
Model using Neural
Networks



Shubham Rawat

Research Interests

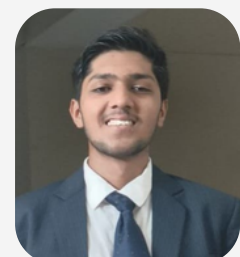
AI/ML Surrogate Modeling,
Autonomous Satellite
Systems.



Paaritosh Jain

Research Interests

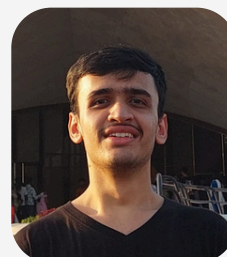
Bayesian inference, ML,
Data analysis, Ray tracing,
MCMC, Propagation of
radio waves in
magnetised plasma.



Aatif Hasan

Research Interests

Radio pulsar population
studies and pulsar
polarization profile
analysis



Prakhar Tripathi

Research Interests

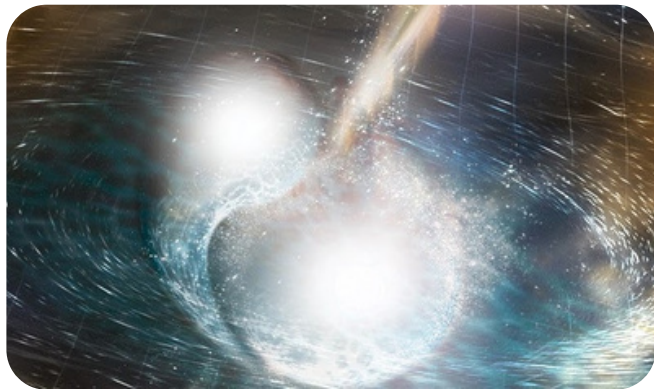
Space Instrumentation,
Detector
Characterization,
Computational
astrophysics

PROJECTS

Astrobiology and Scientific Instrumentation for Planetary Exploration

Faculty Lead: Prof. Yamini

The research focuses on developing scientific instruments and experiments for planetary exploration, astrobiology, and habitability studies. The lab investigates how life responds to extreme planetary-like environments and studies terrestrial analogue sites across the Indian subcontinent and surrounding oceans to identify robust biosignatures. By integrating electrochemistry, spectroscopy, imaging, molecular biology, and bioinformatics, the group develops compact and autonomous space payloads.



Astrophysical Transients & Multi-messenger Astronomy

Faculty Lead: Prof. Kunal P. Mooley

This project encompasses observational and theoretical studies of energetic cataclysmic events like neutron star mergers, supernovae and tidal disruption events that can be observed using two or more messengers, i.e. photons, gravitational waves, neutrinos or cosmic rays.

Fundamental Physics using compact objects

Faculty Lead: Prof. Kunal P. Mooley

Using facilities like the radio very long baseline array, the Hubble Space Telescope and the James Webb Space Telescope, we are obtaining extremely precise positional measurements from neutron stars and black holes to study the physics of strong gravity and derive parameters, like mass, spin and velocity, associated with compact object systems.



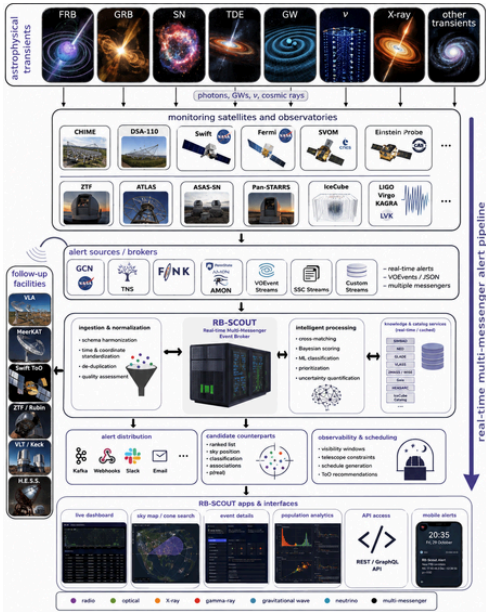
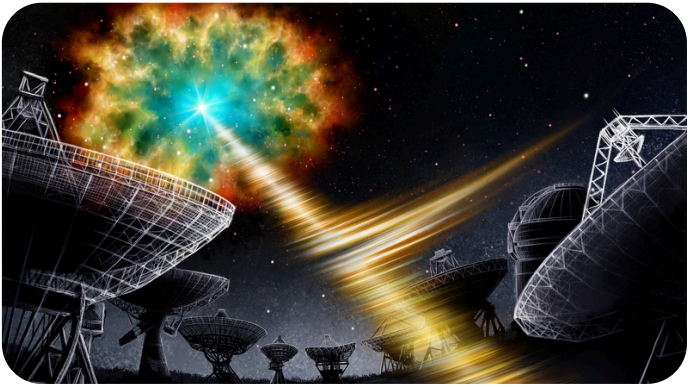
Phased Radio Array for Cosmic-Ray Observations

Faculty Lead: Prof. Pankaj Jain & Prof. Mugundhan V.

This project focuses on developing a phased radio antenna array for detecting radio emissions produced by cosmic-ray air showers. The system is planned for deployment at the GRAPES-3 Cosmic Ray Observatory in Ooty, enabling sensitive and directional observations of high-energy cosmic phenomena.

Bayesian AI Framework for Fast Radio Burst Population Analytics

Faculty Lead: Prof. Mohit Bhardwaj
Develop a scalable Bayesian inference framework to analyze large astronomical datasets and estimate the intrinsic energy distribution and population characteristics of Fast Radio Bursts (FRBs). The project emphasizes statistical modeling, uncertainty quantification, and large-scale data analytics.

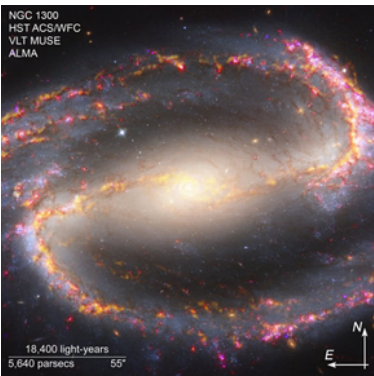
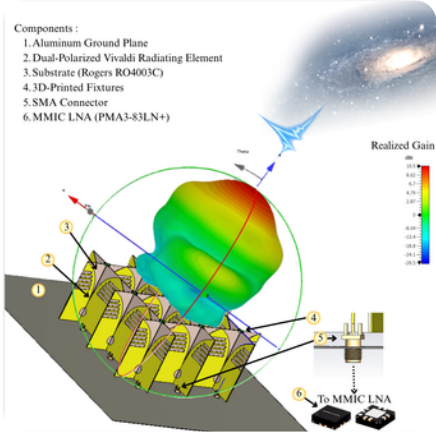


FRB-Scout: A Real-time Multi-Messenger Event Broker for Astronomical Counterpart Discovery

Faculty Lead: Prof. Mohit Bhardwaj
Design and implement an automated real-time multi-messenger astronomical event broker that ingests and normalizes transient alerts from radio, millimeter/sub-mm, infrared (NIR/MIR/FIR), optical, UV, X-ray, and gamma-ray observatories, as well as neutrino and gravitational-wave detectors. The system will perform low-latency spatial and temporal cross-matching of incoming events against archival astronomical catalogs and heterogeneous multi-wavelength datasets, identify potential physical counterparts, characterize their observational and astrophysical properties, and rank candidates using configurable scientific decision pipelines.

GARUDA: Design of a GHz Phased Array for Wide-Field Radio Transient Surveillance

Faculty Lead: Prof. Mohit Bhardwaj
Design a dense phased-array antenna system operating at GHz frequencies for continuous monitoring of the dynamic radio sky and Near-Earth Objects, integrating antenna engineering, beamforming, and array signal processing.

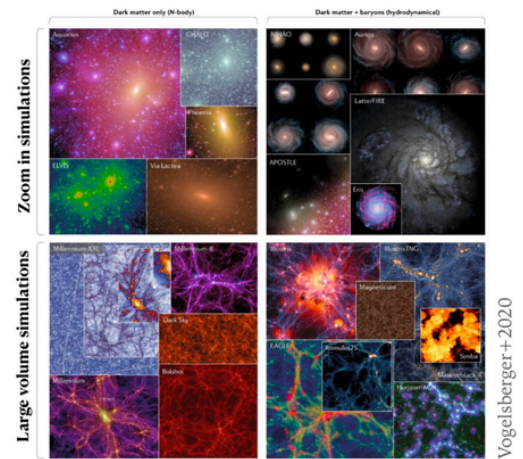


Pattern Speed Evolution of Milky Way Bar

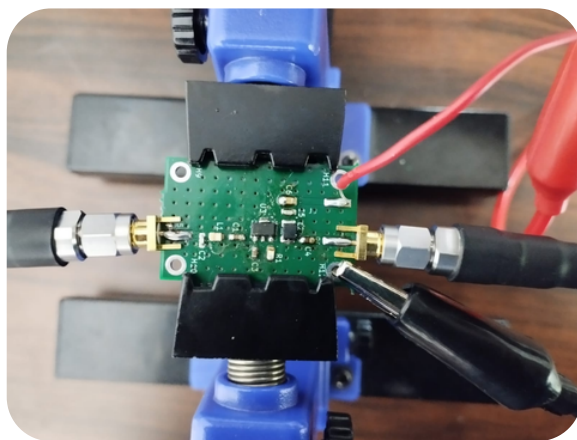
Faculty Lead: Dr. Sandeep Kumar Kataria
Pattern-Speed Evolution of Galactic Bars
This project investigates the discrepancy between observed bar-pattern speeds in nearby galaxies and predictions from EAGLE and IllustrisTNG. It examines sub-grid models to reconcile simulations with observations and understand barred galaxy evolution, including the Milky Way, using the PARAM Supercomputer at IIT Kanpur.

Early Galaxy Formation Frameworks as Predicted by JWST

Faculty Lead: Dr. Sandeep Kumar Kataria
JWST observations reveal galaxies forming earlier than Λ CDM predictions, challenging standard cosmological simulations. This project develops a computational framework for zoom-in simulations to test competing physical processes—star formation, supernovae, black hole feedback, and alternative gravity models like fuzzy dark matter—that may resolve this. Using the PARAM Supercomputer at IIT Kanpur.



Vogelsberger+2020

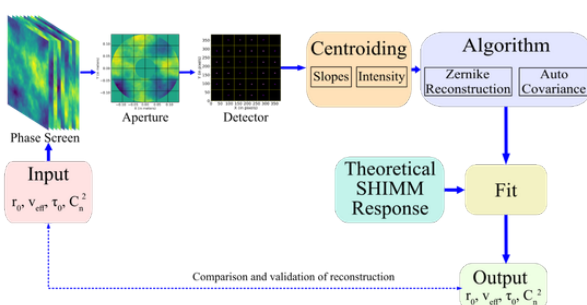
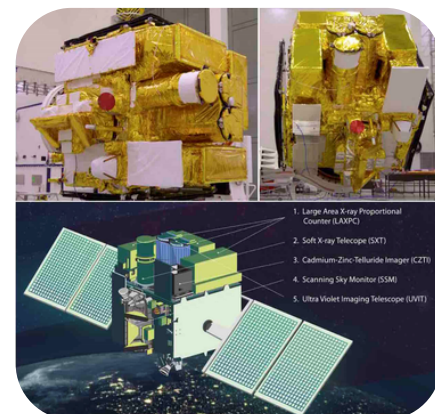


Space Weather & Radio Instrumentation

Faculty Lead: Dr. Mugundhan V.
The Space Instrumentation Group develops specialized payloads for space weather, radio astronomy, and very long baseline interferometry. Its flagship SWARA project, funded by ISRO, develops radio and plasma-wave instrumentation for early warnings of Coronal Mass Ejections and Solar Energetic Particle events, with applications to future planetary missions. The group also develops low-frequency arrays for studying Jupiter's magnetosphere and detecting transient radio emissions from cosmic rays and lightning.

Study of X-ray Binary Star Systems using AstroSat

Faculty Lead: Dr. Pankaj Jain
In collaboration with Prof. J. S. Yadav, I have been working on the X-ray binary star systems using the AstroSat observatory. Here we are interested in understanding the phenomenon of quasi-periodic oscillations seen in these objects.



Atmospheric Characterization for ground-based Optical and near-IR Surveillance.

Faculty Lead: Prof. Prashant Pathak
This research develops a low-cost Shack-Hartmann Image Motion Monitor (SHIMM) to measure atmospheric coherence length, coherence time, and turbulence profiles, supporting telescope site selection and adaptive optics design.

Development of Techniques for Ground-Based Source Localization Using Sparse Array On-board LEO Satellite

Faculty Lead: Prof. Soumyabrata Chakrabarty

Carried out during 30.03.2023 to 29.09.2024 in the Department of Electrical Engineering, IIT Gandhinagar. The project was funded by SAC, ISRO, Ahmedabad under RESPOND of worth Rs. 26,24,520/-

Prediction of Spacecraft Charging Under Real-time Space Environment Conditions

Faculty Lead: Prof. Soumyabrata Chakrabarty

Project approval and funding received from ISRO RESPOND to work in collaboration with URSC, ISRO, Bangalore. Duration of the project: September 2025-August 2027.

Spacecraft Charging in Solar Wind

Faculty Lead: Prof. Rohit Sharma

Spacecraft Charging is the electrical charge buildup on the spacecraft due to the interaction with the space plasma. The electrostatic charge can build up on the surface of the body, as well as inside the volume, known as surface charging and bulk charging effects. Varying plasma environments driven by solar activity lead to varying degrees of charging. This project aims to build simulation tools for surface and bulk currents.

Generative AI for Radiative Transfer

Faculty Lead: Prof. Rohit Sharma

The problem of ray propagation in the solar corona is studied using learning models. The model will learn the propagation process. We use high-resolution solar radio maps and generative ML models to learn low-frequency radio wave propagation in the corona.

PAST WORKSHOP HIGHLIGHTS

SKA-India Summer Training Program

- The department hosted the SKA-India Summer Training Program 2026, a two-week intensive school aimed at developing foundational and practical skills in radio astronomy among advanced undergraduate and postgraduate students. (Indian Institute of Technology Kanpur)
- The programme combined expert lectures, laboratory sessions, and hands-on training in technologies and scientific techniques relevant to the Square Kilometre Array (SKA), one of the world's next-generation radio astronomy facilities. (Indian Institute of Technology Kanpur)



- Participants received exposure to radio instrumentation, observational astronomy, radio astronomical data analysis, high-throughput signal processing, high-performance computing, big-data techniques, and AI/ML applications in astronomy. (Indian Institute of Technology Kanpur)
- A key practical component involved hands-on construction of a radio telescope, giving participants experience with the instrumentation underlying modern radio astronomy. (Indian Institute of Technology Kanpur)
- Scientific sessions covered topics including pulsars, cosmology, Galactic and extragalactic astronomy, radiation mechanisms, radio telescopes, and observational techniques. (Indian Institute of Technology Kanpur)

- The programme brought together experts from IIT Kanpur, IIT Indore, IIST, IUCAA, NCRA-TIFR, BITS Goa, MAHE and other institutions, providing participants with broad exposure to India's radio astronomy community. (Indian Institute of Technology Kanpur)



- Laboratory sessions were conducted within the Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE), IIT Kanpur, highlighting the department's facilities and growing capabilities in radio astronomy and instrumentation. (Indian Institute of Technology Kanpur)

Anslys STK Simulation Challenge & Workshop

- The department hosted a two-day hands-on Ansys STK Simulation Challenge & Workshop on 15–16 April 2026, providing students with practical exposure to simulation-driven space mission engineering.
- Participants worked with Ansys Systems Tool Kit (STK) on real-world engineering problems, gaining experience in modelling, analysis, and mission simulation.
- Technical sessions covered space mission design, orbital mechanics, sensor modelling, access and coverage analysis, and mission analysis using Astrogator.
- The workshop also introduced Python-based automation in STK, exposing students to computational workflows for space-system analysis.
- A team-based simulation challenge enabled participants to apply the concepts learned during the workshop and present their solutions for evaluation.
- The event featured expert-led technical sessions by CADFEM India, strengthening students' exposure to industry-standard simulation tools and engineering practices.

CONTACT US

Student's Placement Office
109, Outreach Building, IIT Kanpur,
Kanpur-208016
Phone no.: +91 512 259 44 33/34
Email: spo@iitk.ac.in



Prof. Mohit Bhardwaj

Faculty Placement Coordinator

202-c, Western Extension Lab, IIT Kanpur

Email: mohitb@iitk.ac.in

Contact No. **+91-512-259-2641**

Website: <https://iitk.ac.in/mohit-bhardwaj>



Intakhab Alam

Department Placement Coordinator

Email: intakhab25@iitk.ac.in

Contact No. **+91 8448138448**

LinkedIn: <https://www.linkedin.com/in/intakhab08/>